

Examiner
only

9. The following table shows the decomposition temperatures of Group 2 metal carbonates and nitrates.

Metal	Temperature at which the carbonate decomposes (°C)	Temperature at which the nitrate decomposes (°C)
magnesium	117	89
calcium	178	561
strontium	235	570
barium	267	700

(a) Describe the trends in the stabilities of the Group 2 carbonates and nitrates. [3]

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(b) When a carbonate decomposes it produces carbon dioxide gas. Describe an experiment that could be carried out to show that carbon dioxide gas is produced during the reaction. [2]

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(c) When calcium nitrate decomposes it forms calcium oxide, oxygen and nitrogen dioxide, NO₂.

Write a **symbol** equation for the reaction. [3]

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